

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) (a) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$, $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does **not** hold for this $\{A_n\} \subset \mathcal{F}$.

(b) On the same $(\Omega, \mathcal{F})$ and with the same $\{A_n\} \subset \mathcal{F}$, can you specify a *different* measure μ' such that the monotone continuity from above does hold this $\{A_n\} \subset \mathcal{F}$?

(c) In general, under what conditions does the monotone continuity from above hold?

(d) In general, under what conditions does the monotone continuity from *below* hold? (4 × 10 = 40 points)

(2) Let $\mathcal{F}$ be a σ -algebra of subsets of $\Omega \neq \emptyset$. Take $\Omega_0 \subset \Omega$ such that $\Omega_0 \notin \mathcal{F}$. Define $\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

(15 points)

(3) (a) Give an example of a function f that is **not** $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(b) Give an example of a function f that is **not** $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable, but $|f|$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(2 × 10 = 20 points)

(4) (a) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, (-\infty, 5], (5, \infty), \mathbb{R}\}$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable? Justify your answer.

(b) Let $\{\epsilon_i\}_{i \geq 1}$ be a sequence of *random variables* (i.e. measurable functions) defined on some probability space $(\Omega, \mathcal{F}, P)$ and $\{x_i\}_{i \geq 1}$ be a sequence of real numbers (with $\sum x_i^2 > 0$). For some $\beta \in \mathbb{R}$, define a new sequence of real valued functions on $(\Omega, \mathcal{F}, P)$:

$$Y_i(\omega) = x_i \beta + \epsilon_i(\omega), \quad \omega \in \Omega, \quad i \geq 1.$$

and using the sequence $\{Y_i\}_{i \geq 1}$, define the following ~~two~~ real valued functions on $(\Omega, \mathcal{F}, P)$ for each $n \geq 1$

$$\hat{\beta}_n(\omega) = \frac{\sum_{i=1}^n x_i Y_i(\omega)}{\sum_{i=1}^n x_i^2}, \quad \omega \in \Omega.$$

Argue that $\{Y_i : i \geq 1\}$, $\{\hat{\beta}_n, n \geq 1\}$ are sequences of real valued *measurable* functions on $(\Omega, \mathcal{F}, P)$ (i.e. random variables).

(2 × 15 = 30 points)

1. (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $\mu(A_n) = \infty$, $\bigcap_{n \geq 1} A_n = \emptyset$. So $\mu(\bigcap_{n \geq 1} A_n) \neq \lim_{n \rightarrow \infty} \mu(A_n)$. Hence m.c.f.a doesn't hold.
- (b) Take $\mu'(A) = \sum_{i \in A} \frac{1}{2^i}$. Then $\mu(\bigcap_{n \geq 1} A_n) = 0 = \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} = \lim_{n \rightarrow \infty} \mu(A_n)$. So m.c.f.a holds.
- (c) If $\exists n_0$ s.t $\mu(A_{n_0}) < \infty$, then m.c.f.a holds.
- (d) Always true.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\bigcup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\bigcup_{i \geq 1} A_i = (\bigcup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

3. (a) $f = 1_A$, where A is a non-Borel measurable set.

- (b) $g = 1_A - 1_{A^c} = \begin{cases} 1; & \text{on } A \\ -1; & \text{on } A^c \end{cases}$ is not measurable, but $|f| \equiv 1$ is Borel measurable (constant functions are always measurable).

5. (a) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable.

- (b) Note that $\{\epsilon_i : i \geq 1\}$ are measurable, and for each $i \geq 1$, Y_i is created by adding a constant to ϵ_i and hence Y_i is measurable. For each $n \geq 1$, $\hat{\beta}_n$ is created by multiplication, component-wise addition and division (multiplication) by a constant of measurable functions $\{Y_i : i \geq 1\}$ and hence they are measurable too.

① (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. $A_n = \{j: j \geq n\}$ $n \geq 1$ $\mu(A) = |A|$.

Want to show that MCFA does not hold for this.

Consider that

$$A_n \supset A_{n+1}$$

MCFA says that

$$\lim_{n \rightarrow \infty} \mu(A_n) = \mu\left(\bigcap_{n=1}^{\infty} A_n\right)$$

In this case

$$\mu(A_n) = |A_n| = \infty \quad \forall n \geq 1 \quad \text{so} \quad \lim_{n \rightarrow \infty} \mu(A_n) = \infty$$

and

$$\begin{aligned} \mu\left(\bigcap_{n=1}^{\infty} A_n\right) &= \mu\left(\left(\left(\bigcap_{n=1}^{\infty} A_n\right)^c\right)^c\right) = \mu\left(\left(\bigcup_{n=1}^{\infty} A_n^c\right)^c\right) \\ &= \mu\left(\left(\bigcup_{n=1}^{\infty} \{1, 2, \dots, n\}\right)^c\right) \\ &= \mu\left(\mathbb{N}^c\right) \\ &= \mu(\emptyset) \\ &= |\emptyset| \\ &= 0 \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} \mu(A_n) = \infty \neq 0 = \mu\left(\bigcap_{n=1}^{\infty} A_n\right). \quad \checkmark$$

(b) We need to find some μ' s.t. $\mu'(A_n) < \infty$ for $n \geq 1$.

Let $\mu(A) = 0 \quad \forall A \in \mathcal{F}$. Then μ is a measure

since

- (i) $\mu(\emptyset) = 0$
- (ii) $\mu(A) \in [0, \infty] \quad \forall A \in \mathcal{F}$
- (iii) ...

(c) MCFA holds where $\exists$ some $n_0 \geq 1$ s.t. $\mu(A_{n_0}) < \infty$
for $A_{n_0} \in \{A_n\}_{n \geq 1} \in \mathcal{F}$

(d) MCFB always holds

(2) $\mathcal{F}$ a σ -algebra of subsets of $\Omega \neq \emptyset$. $\Omega_0 \subset \Omega$ s.t. $\Omega_0 \notin \mathcal{F}$.

$$\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$$

We want to show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 . Consider $F \cap \Omega_0 \in \mathcal{P}(\Omega_0)$ since $\Omega_0 \notin \mathcal{F}$.

(i) Need to show $\Omega_0 \in \mathcal{F}_0$. Consider that

$$\Omega \cap \Omega_0 \in \mathcal{F}_0 \text{ since } \Omega \in \mathcal{F} \text{ since } \mathcal{F} \text{ a } \sigma\text{-alg.}$$

$$\Rightarrow \Omega \cap \Omega_0 = \Omega_0 \in \mathcal{F}_0 \quad \checkmark$$

(ii) Let $A \in \mathcal{F}_0$. Need to show $A^c \in \mathcal{F}_0$.

$$\text{Consider } A^c = \Omega_0 \setminus A = \Omega_0 \cap A^c \text{ where}$$

$$A^c = (F \cap \Omega_0)^c \text{ for some } F \in \mathcal{F}$$

$$= F^c \cup \Omega_0^c$$

$$= F^c \cup \emptyset$$

$$= F^c$$

$$\Rightarrow A^c = \Omega_0 \cap A^c = \Omega_0 \cap F^c \in \mathcal{F}_0$$

since $F^c \in \mathcal{F}$ since $\mathcal{F}$ a σ -algebra. $\checkmark$

(iii) Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_0$. Need to show $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$

$$\bigcup_{n \geq 1} A_n = \bigcup_{n \geq 1} (F_n \cap \Omega_0) = \left(\bigcup_{n \geq 1} F_n \right) \cap \Omega_0 \in \mathcal{F}_0$$

since $\left(\bigcup_{n \geq 1} F_n \right) \in \mathcal{F}$ since $\mathcal{F}$ a σ -algebra.

③ (a) Let A be a non-Borel measurable set. Then

let

$$f = I_A.$$

f is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(b) Let

$$f = I_A - I_{A^c} = \begin{cases} 1 & \text{if } x \in A \\ -1 & \text{if } x \notin A \end{cases}$$

but $|f| = 1 \quad \forall x$, so $|f|$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

④ (a) Consider that for $A \in \mathcal{F}_2$,

$$T^{-1}(A) = \begin{cases} \emptyset & \text{if } 0 \notin A, 5 \notin A \\ (-\infty, 0] & \text{if } 0 \notin A, 5 \in A \\ (0, \infty) & \text{if } 0 \in A, 5 \notin A \\ \mathbb{R} & \text{if } 0 \in A, 5 \in A. \end{cases}$$

Now $(-\infty, 0]$ and $(0, \infty)$ are not in $\mathcal{F}_1$, so T is not $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable.

(b) Let us first argue that Y_i is measurable for all $i \geq 1$. We know that E_i is measurable for $i \geq 1$,

and x_i/β is a constant in $\mathbb{R}$, so Y_i is equal to the summation of a constant (which is always measurable) and a measurable function. Hence, Y_i is measurable for all $i \geq 1$. Now consider

$\hat{\beta}_n$ for $n \geq 1$.

- $x_i y_i$ is the product of two measurable functions, so measurable
- $\sum_{i=1}^n x_i y_i$ is the summation of measurable functions, so measurable
- $x_i^2 = x_i \cdot x_i$ is the product of two measurable functions, so measurable
- $\sum_{i=1}^n x_i^2$ is the summation of measurable functions, so measurable
- $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \left(\sum_{i=1}^n x_i y_i \right) \left(\sum_{i=1}^n x_i^2 \right)^{-1}$, so it is the product of measurable functions and is therefore measurable.
 $\swarrow$ also measurable

Hence, $\hat{\beta}_n$ is measurable for $n \geq 1$.